

SOA Survey

Site-Invariant Alzheimer's Detection via Bio-Acquisition Latent Factorization

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Abstract

The detection of Alzheimer's disease (AD) using neuroimaging is a critical problem in Alzheimer's disease research. Early and accurate detection of Alzheimer's disease can be helpful in timely interventions and prognosis. However, deep learning-based approaches using MRI and PET scans often suffer from performance degradation when tested using different data sets. In this research survey, we will examine this problem from a robust multimodal representation learning perspective. In this research survey, we reviewed nine research papers related to multimodal self-supervised learning, multimodal domain adaptation, multimodal domain generalization, image harmonization, multimodal fusion, and rigor in Alzheimer's disease imaging research. From this literature research, it is evident that multimodal MRI-PET and site-invariant learning have been studied. However, acquisition shifts are still coarsely addressed and not rigorously checked if scanner information is still present in the learned disease representation. Based on this literature research gap, we will explore a disease-acquisition latent factorization approach that will allow researchers to explicitly check if robustness is present in Alzheimer's disease imaging research.

1. Problem Definition

The detection of Alzheimer's disease using neuroimaging is often posed as a classification problem in which a model is trained to classify a subject into a class like cognitively normal (CN), mild cognitive impairment (MCI), Alzheimer's disease (AD) and so forth using structural MRI scans and/or PET scans. In the context most relevant to our project, a paired MRI-PET dataset is provided as input to a model that learns a representation that is strong enough to support a diagnosis of Alzheimer's disease. This is a problem that is important because structural MRI scans can provide information that is useful for a diag-

nosis of Alzheimer's disease, and PET scans can provide functional information like glucose metabolism and amyloid deposition. A model that can leverage this information from two different sources can potentially leverage more information than a model that is restricted to a single imaging modality [1, 5, 7].

The problem is that Alzheimer's imaging datasets are acquired at many different institutions and using many different scanners and imaging protocols. Deep networks can leverage these shortcuts rather than the underlying biology of Alzheimer's disease and can degrade in performance when tested on a new scanner or new cohort [2, 3, 8]. In Alzheimer's disease in particular, many studies report excellent performance internally but poor external validation and poor control of confounds [8]. From a deep learning perspective, this is a difficult problem for a number of reasons. It is a difficult problem because Alzheimer's imaging datasets are heterogeneous and often restricted by access and permission; it is a difficult problem because Alzheimer's imaging datasets are often paired between imaging modalities; it is a difficult problem because deep network design is a difficult problem; and it is a difficult problem because evaluation is a difficult problem.

2. Literature Review

2.1. Generalization under site, scanner, and cohort shift

One of the biggest challenges in AD imaging is that a model trained on one cohort often fails to generalize well to another cohort since acquisition conditions vary across sites. In their recent paper, Fu et al. directly address this problem using a multi-modal deep

domain adaptation approach that aligns MRI and PET feature spaces across domains and retains diagnostic information [2]. While this paper is useful in that it demonstrates that some level of performance can be recovered even across domains using a multi-modal approach, the basic idea is still alignment-based rather than factorization-based. In another recent paper, Lteif et al. take a different perspective on the problem of domain generalization and contend that a model should be steered toward patterns that are relevant to disease and transfer across cohorts like ADNI, FHS, AIBL, and NACC [3]. While this is relevant to our project in that it takes a more robust perspective rather than a training trick, these papers all demonstrate that generalization across domains is a well-recognized bottleneck.

However, what these papers leave unanswered is a more precise question: how can we be sure that a certain part of the learned representation is related to disease and another part is related to acquisition? While many approaches reduce domain mismatch across all dimensions, none of these approaches prove that scanner information has been removed from the latent space. In a multimodal setting like AD imaging, this is particularly problematic since each modality has its own acquisition artifacts.

2.2. Multimodal fusion and representation learning

The second challenge is how to learn useful multimodal representations from MRI and PET images without reducing one modality to another or overfitting to data-set-specific information. The paper that is closest to our desired direction is by Ali et al., where they propose multimodal self-supervised learning for early AD, including cross-modal alignment of MRI and PET, longitudinal consistency, and optimization of site-invariant embedding. Their results demonstrate that multimodal pre-training can be useful for downstream diagnosis, and that site-aware robustness is possible. Li et al. introduce DiaMond, a multimodal vision transformer that uses self-attention, bi-attention, and multimodal normalization for MRI and PET fusion, demonstrating that a well-designed multimodal model can improve diagnostic performance. Fedorov et al., although not specifically focused on

AD, demonstrate that self-supervised multimodal neuroimaging can learn useful information without relying on labels, including regions that are relevant for disorders and latent links.

This line of research is very relevant to our own, since it demonstrates that multimodal representation learning is a good direction. However, the main focus of all of these papers is on how to better fuse, how to better self-supervise, or how to better learn multimodal embeddings. What is missing is a principled way of separating the latent space of MRI/PET images into disease factors and acquisition factors, and then validating that separation against external data.

2.3. Harmonization, methodological rigor, and what still remains unresolved

The third challenge is to determine whether to resolve the variability in image acquisition through image harmonization or learning invariant representations or a combination of both. Choi et al. demonstrated the efficacy of harmonizing amyloid PET images using deep learning techniques in predicting cognitive decline, making harmonization a potential approach to address variability in image acquisition [4]. Zuo et al. also proposed a harmonization technique called HACA3 to harmonize MRI images acquired from multiple sites, which can handle contrast agent variability, anatomical preservation, and robustness to image artifacts [6]. The importance of these papers lies in the demonstration that variability in image acquisition can be addressed at the image level by directly dealing with scanner-tracer mismatches. However, image harmonization might not ensure the complete absence of disease embeddings in the learned representation due to the presence of acquisition signal in the input.

The quality of evaluation is also a key challenge in the field, as demonstrated in the recent survey by Guan et al., which organized the multimodal prediction techniques for AD and pointed to the outstanding challenges in the availability of data, modality imbalance, realism, and generalization [7]. Young et al. took a more critical view of the field by demonstrating the widespread existence of data leakage in AD deep learning models due to poor validation practices [8]. This indicates that any proposed solution in the field needs to be validated not only in terms of accuracy

but also in terms of subject-wise splitting, testing, and evaluation for potential confounding effects.

Open Problems (flagged by the field:)

- External generalization still has to be improved, particularly in scenarios where training and test conditions are significantly different [3, 7].
- Fusion approaches still need to improve their robustness to acquisition variations, modality imbalances, and realistic clinical scenarios [1, 7].
- Methodological rigor is still a limiting factor; subject leakage and performance inflation by confounds are still issues in AD deep learning research [8].

3. Research Gap and Proposed Direction

The literature reviewed above indicates that this research space has already explored three close concepts: multimodal MRI-PET fusion, multimodal self-supervised representation learning, and domain adaptation or site-invariant training. This implies that our research contribution is not simply “use MRI and PET together and reduce site effects.” This space has already been touched by recent research, particularly by Ali et al. and Fu et al. [1, 2]. However, it is worth noting that this space has a gap. Most of the current research has aligned domains, harmonized images, or improved multimodal fusion modules. Yet, none of these works has explicitly modeled acquisition nuisance as a structured latent factor separate from disease biology. Furthermore, none of them has validated that this deconfounding is successful by checking for residual leakage.

The gap in this research space that we are choosing is: “previous works improve invariance to shift, but do not cleanly separate *what* is being made invariant.” For our task, this is a distinction that is important. Acquisition nuisance is hierarchical and multimodal. For instance, factors like site, scanner vendor, strength, PET tracer, and reconstruction protocol all may be relevant for model behavior. For our task, MRI and PET do not inherit these effects in a similar manner. Therefore, we propose that we investigate a disease-acquisition latent factorization, in which a single latent is learned for disease biology for di-

agnosis, and separate acquisition latents are learned for nuisance factors. What we hope to investigate is whether this latent factorization improves, not simply out-of-domain performance, but whether it improves this performance by having lower residual scanner leakage than fusion, harmonization, or adversarial alignment.

Research Question (formal: 1–3 lines): Can a multimodal MRI–PET representation for Alzheimer’s disease be decomposed into disease-relevant and acquisition-relevant latent factors such that AD diagnosis remains accurate while residual information about site, scanner, field strength, tracer, and protocol is reduced under external evaluation?

4. Benchmark, Evaluation, and Data Landscape

Datasets. The core dataset infrastructure primarily revolves around the multi-site AD cohorts with longitudinal MRI and PET imaging, though the access to the imaging tends to be limited rather than open. ADNI remains the most common development dataset, and it has longitudinal clinical and biomarker information for over 2,500 participants, with MRI and PET imaging information accessible through the LONI archive. OASIS-3 also provides a valuable external dataset with 1,378 participants and multimodal MRI and PET imaging information. AIBL provides another valuable external dataset, with the full dataset comprising more than 3,000 participants, though the imaging dataset is smaller in size and also mediated through the infrastructure of the study. NACC provides a valuable dataset for broader external validation, with 54,000+ participants overall and 15,000+ participants with MRI and PET imaging information, though it tends to be more heterogeneous and less standardized for paired experiments [3, 4, 5].

Evaluation Protocol. The most frequently used evaluation metrics in this field are accuracy, balanced accuracy, sensitivity, specificity, F1-score, and AUC. Balanced accuracy and AUC are particularly relevant for this problem because of class imbalance. Moreover, robustness work should not obscure poor performance on the minority class using accuracy [5, 2].

Since this work is fundamentally about confound-resistant representations, we should also consider subject-wise splitting, external testing cohorts, worst-group performance across acquisition strata, and linear leakage probes that attempt to learn site or scanner information from the disease latent [8, 3]; this is more appropriate for the task than a single accuracy figure over internal testing.

Compute Context. The majority of the papers surveyed make use of GPU-based training for 3D models in neuroimaging. This is particularly true when using transformers, diffusion models, and multimodal self-supervised learning [1, 5]. A student-scale implementation of this work should therefore assume access to at least one such device. Moreover, this initial work should restrict itself to paired MRI-PET classification and a limited number of baselines.

Experimental Design Considerations

A rigorous experiment for this direction would involve comparing against baselines that capture the primary alternative explanations for improved robustness. First, we would need a multimodal fusion baseline without deconfounding, i.e., an MRI-PET classifier without any deconfounding step. This is because we want to know if any improvement is merely due to multimodal fusion. Second, we would need a domain adaptation adversarial invariance baseline inspired by Fu et al. and Ali et al. This is because we want to know if deconfounding is better than making the whole embedding invariant. Third, we would need a harmonization baseline that is conceptually inspired by HACA3 and PET harmonization work. This is because we want to know if pre-processing level correction is better than deconfounding.

The ablations that we would need to perform to make this work meaningful are ablations that involve deconstructing the various parts of this idea rather than merely tuning hyperparameters. One ablation that we would need to perform is comparing a single latent space against a split between disease and acquisition. A second ablation that we would need to perform is whether acquisition should be modeled as one generic factor versus whether acquisition should be modeled as a hierarchy of site, scanner, field strength, tracer,

and protocol. A third ablation that we would need to perform is whether we should merely choose models based on mean validation accuracy. In fact, we know that there are several evaluation pitfalls that we must avoid. These include subject leakage between train and test sets, where scans from the same individual appear in multiple sets. In addition, we also know that we must avoid metric inflation when testing on single cohorts. Moreover, we also know that we must avoid overclaiming interpretability without demonstrating robustness. A rigorous evaluation would therefore involve using subject-wise splits and a held-out external validation set.

5. Annotated Reading List

References

- [1] S. B. Ali, H. Ghannam, I. Yakushev, D. M. Hedderich, and C. Wachinger. Multimodal Self-Supervised Learning for Early Alzheimer’s: Cross-Modal MRI-PET, Longitudinal Signals, and Site Invariance. *Diagnostics*, 2025. *What it does: The paper proposes a multimodal self-supervised framework that jointly learns MRI and PET features by employing cross-modal alignment, longitudinal consistency, and site-invariant embedding optimization. What it contributes: The paper demonstrates that multimodal pre-training is beneficial for AD-related diagnosis, addressing the issue of site variation in the representation space. Why it is here: This is the best prior work in our direction, and it defines the strongest of the nearby baselines for robust multimodal representation learning.*
- [2] B. Fu, et al. Prediction of Alzheimer’s Disease Based on Multi-Modal Domain Adaptation. *Brain Sciences*, 2025. *What it does: builds a multi-modal deep domain adaptation model to classify AD under domain mismatch using MRI and PET. What it contributes: demonstrates that aligning multimodal features across domains can improve performance under distribution shift. Why it is here: it is a key baseline family paper for our project because our method must show value beyond standard domain adaptation.*

- [3] D. Lteif, et al. Disease-driven domain generalization for neuroimaging-based assessment of Alzheimer’s disease. *What it does*: studies domain generalization for AD neuroimaging by encouraging the model to concentrate on disease-related signal that transfers across cohorts. *What it contributes*: presents evidence that representation design is important for external robustness, not just in-domain performance. *Why it is here*: it relates to the OOD aspect of our project and demonstrates the importance of cross-cohort evaluation instead of internal evaluation.
- [4] Y. S. Choi, et al. Deep learning-based amyloid PET harmonization to predict cognitive decline in non-demented elderly. *Radiology Advances*, 2024. *What it does*: learns tracer-robust amyloid PET representations through harmonization and uses them for downstream cognitive decline prediction. *What it contributes*: shows that acquisition harmonization across PET tracers can improve clinically relevant downstream prediction. *Why it is here*: it is directly relevant to our nuisance-shift problem and represents a strong image-level harmonization reference.
- [5] Y. Li, M. Ghahremani, Y. Wally, and C. Wachinger. DiaMond: Dementia Diagnosis with Multi-Modal Vision Transformers Using MRI and PET. *Proceedings of WACV*, 2025. *What it does*: introduces a multimodal vision transformer with self-attention, bi-attention, and multimodal normalization for MRI-PET dementia diagnosis. *What it contributes*: offers a state-of-the-art multimodal fusion technique with promising results in AD diagnosis and differential diagnosis. *Why it is here*: it follows the fusion-centric thread that our work needs to be compared against in both philosophy and experiments.
- [6] L. Zuo, et al. HACA3: A unified approach for multi-site MR image harmonization. *Computerized Medical Imaging and Graphics*, 2023. *What it does*: proposes a harmonization technique that separately accounts for the effects of contrast, anatomy, and artifacts in MRI images for harmonization. *What it contributes*: demonstrates the possibility of making harmonization more flexible in the choice of MRI contrasts and more robust in the presence of different types of artifacts. *Why it is here*: it provides us with a state-of-the-art harmonization-based technique that we compare against in the absence of disease and acquisition factorization.
- [7] Y. Guan, W. Wang, J. Chen, P. Yang, J. Xu, and J. Qi. A Survey of Multimodal Fusion for Alzheimer’s Disease Prediction: A New Taxonomy and Trends. *Information Fusion*, 2025. *What it does*: Reviews AD deep learning studies using a lens of data leakage, validation rigor, and performance inflation. *What it contributes*: Demonstrates that inflated accuracy results are not methodologically rigorous when considering data leakage and validation. *Why it is here*: Directly informs our evaluation approach and robustness requirements.
- [8] V. M. Young, S. Gates, L. Y. Garcia, and A. Salardini. Data Leakage in Deep Learning for Alzheimer’s Disease Diagnosis: A Scoping Review of Methodological Rigor and Performance Inflation. *Diagnostics*, 2025. *What it does*: reviews AD deep learning papers through the lens of data leakage, validation rigor, and performance inflation. *What it contributes*: shows that many high reported accuracies are not methodologically trustworthy when leakage and weak validation are accounted for. *Why it is here*: it directly informs our evaluation design and the need for leakage-audited robustness claims.
- [9] A. Fedorov, E. Geenjaer, L. Wu, and V. Calhoun. Self-supervised multimodal learning for group inferences from MRI data: Discovering disorder-relevant brain regions and multimodal links. *NeuroImage*, 2024. *What it does*: Introduces a novel framework for self-supervised multimodal learning that can learn multiple representations without labels. *What it contributes*: Demonstrates that multimodal learning can discover informative disorder-linked regions and latent relations. *Why it is here*: Provides support for multimodal representation learning that underpins our work, although not specific to MRI-PET AD diagnosis.

6. Preliminary Implementation Resources

- <https://github.com/ai-med/DiaMond> Official PyTorch code for the DiaMond WACV 2025 paper; useful as a direct MRI–PET fusion baseline and as a reference for paired HDF5-based multimodal training.
- https://github.com/Kateridge/TransMF_AD Transformer-based multimodal fusion code for early AD diagnosis; helpful for understanding MRI/PET fusion pipelines and for comparing against a simpler transformer baseline.
- <https://github.com/Kateridge/MultimodalAD> Code for a modality-flexible AD diagnosis framework; useful if later deliverables require experiments under incomplete or clinically variable modality availability.
- <https://github.com/ai-med/PASTA> Official implementation of a pathology-aware MRI-to-PET diffusion model; relevant as a cross-modal learning reference and for ideas around MRI/PET consistency.
- <https://github.com/lianruizuo/haca3> Official repository for HACA3 harmonization; relevant as an implementation reference for multi-site MRI harmonization and nuisance-robust preprocessing.
- <https://github.com/aramis-lab/AD-ML> A reproducible framework for AD classification with MRI and PET on ADNI, AIBL, and OASIS; very useful for benchmarking and for understanding multi-cohort experimental setup.
- <https://github.com/hayleyshanks/Longitudinal-MRI-PET-preproc> Longitudinal MRI/PET preprocessing code; useful for understanding practical preprocessing decisions if we build our own paired MRI–PET pipeline.

Table 1: Key datasets in the field.

Dataset	Size	Modality	Notes
ADNI	>2.5k participants	MRI, PET, clinical	Most common benchmark; access-controlled via LONI; strong for development and paired MRI–PET.
OASIS-3	1,378 participants	MRI, PET, clinical	Strong external test cohort; longitudinal and multimodal.
AIBL	3,045 total cohort	MRI, PET, clinical	Important external cohort; imaging subset is smaller than full cohort; access-controlled.
NACC	54k+ total; 15k+ MRI/PET	MRI, PET, clinical	Useful for external validation; highly heterogeneous multi-center data.
J-ADNI	cohort-scale	MRI, PET, clinical	Often used with ADNI for cross-dataset testing; access required.
FHS	cohort-scale	MRI, clinical	Appears in DG work; useful for broader generalization analysis.